



Illuminating
ENGINEERING SOCIETY

APPROVED METHOD:
DETERMINATION OF AVERAGE
LUMINANCE (CALCULATED) FOR
INDOOR LUMINAIRES
AN AMERICAN NATIONAL STANDARD



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has been approved by IES.
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should be directed to IES.

**Prepared by the
IES Testing Procedures Committee**



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1.0 Introduction and Scope

1.1 Introduction

The concept and limitations of average luminance is addressed in this document. Although simple projected area examples are presented and more detailed area calculation methods are developed for reference purposes in **Annex A**, it remains the user's responsibility to seek out the most appropriate methods or formulas each time he or she determines the actual projected areas for a specific luminaire.

1.2 Scope

The methods of calculating average luminance, contained in this Guide, cover various open bottom apertures as well as flat and drop lensed units including units, with multiple openings in the light emitting area. The candela values of interest are obtained by means of IES techniques for relative or absolute photometry, and are not obtained from field measurements in application. *Note:* Average luminaire luminance is not a reliable indicator of either direct or reflected glare due to potential luminance non-uniformity.

The averaging of data from spot luminance measurements obtained with luminance meters or high resolution imaging systems is not within the scope of this document. This calculation is based solely on the candela values obtained by goniometric measurement of the luminaire and the luminous projected area.

All light sources, for which there are current standards for luminaire photometry, are covered by this document. These include incandescent, fluorescent, high intensity discharge (HID), low pressure sodium, and LED sources.

2.0 Normative References

Illuminating Engineering Society. *ANSI/IES LS-1-20, Nomenclature and Definitions for Illuminating Engineering*. New York: IES; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 Jan 11).

3.0 Nomenclature and Definitions

3.1 AB

The luminous area of the bottom of a lens case, measured in square meters (preferred) or square feet, viewed from nadir.

3.2 ABP

The projected luminous area of the bottom of a lens case as viewed from angle θ from nadir, measured in square meters (preferred) or square feet.

3.3 angle θ (vertical viewing angle)

This angle is expressed in degrees as measured from nadir. It is the angle at which the area projections are calculated, and at which the appropriate luminaire intensity (cd) is selected for average luminance calculations.

3.4 angle ψ (horizontal viewing angle)

The lateral position from a beginning reference position, from which performance information is evaluated.

3.5 AS

The luminous area of the side of a drop-lens case, measured in square meters (preferred) or square feet, viewed from 90 degrees vertical.

3.6 ASP

The projected luminous area of the side of a drop-lens case, measured in square meters (preferred) or square feet, as viewed from angle θ from 90 degrees vertical.

3.7 average luminance (calculated)

The average luminance of a luminaire based on the candela value obtained from a goniometric measurement and the projected luminous area of the luminaire at that measurement angle.

$$\text{Average luminance (calculated)} = \frac{\text{luminous intensity at angle } \theta}{\text{projected luminous area at angle } \theta} \quad (3-1)$$

3.8 D

The luminaire lens or opening diameter in meters (preferred) or feet.

3.9 H

The luminaire lens height in meters (preferred) or feet.

3.10 L, W

The rectangular luminaire lens or opening dimension (length, width), expressed in meters (preferred) or feet.

3.11 total projected luminous area (at viewing angle θ)

The sum of the bottom and side luminous projected areas (for a given luminaire lens or opening) as viewed from angle θ at a given horizontal angle ψ .

4.0 General Method of Calculation of Average Luminance (Calculated)

The equation used to determine average luminance as used in this document is:

$$\text{Average luminance (calculated)} = \frac{\text{luminous intensity at angle } \theta}{\text{projected luminous area at angle } \theta} \quad (4-1)$$

where:

angle θ is the viewing angle in degrees (see **Section 3 Nomenclature and Definitions**)

luminous intensity at angle θ is in candelas

projected luminous area at angle θ is in m²

average luminance (calculated) is expressed in cd/m²

Metric (SI) units (as shown for **Equation 4-1**) are preferred, although it is recognized that individual circumstances may require that the equation be implemented using United States Customary System (USCS) units:

$$\text{Average luminance (calculated)} = \frac{\text{luminous intensity at angle } \theta}{\text{projected luminous area at angle } \theta} \quad (4-2)$$

where:

angle θ is the viewing angle in degrees (see **Section 3 Nomenclature and Definitions**)

luminous intensity at angle θ is in candelas

projected luminous area at angle θ is in ft²

average luminance (calculated) is expressed in candelas/ft²

Note: Earlier methods, and even some of today's literature, express luminance in footlamberts (fL), a Lambertian unit of luminance that equals 1/π cd/ft². This terminology is obsolete and actually represents a π-multiple of luminance as calculated in USCS units. It is often presented as:

$$\text{Average "luminance" (calculated)} = \frac{\pi(\text{luminous intensity at angle } \theta)}{\text{projected luminous area at angle } \theta} \quad (4-3)$$

where:

angle θ is the viewing angle in degrees (see **Section 3 Nomenclature and Definitions**)

luminous intensity at angle θ is in candelas

projected luminous area at angle θ is in ft²

average "luminance" is in footlamberts

The examples immediately following in **Sections 4.1** through **4.6** are useful to clarify the quantities involved, giving examples of projections of luminous areas of luminaires. (For more complex shapes, please refer to **Annex A** for calculation methods.)

Caution: These calculated average luminance results are best suited to describe emitter sources that are uniformly diffuse; they do not accurately convey the comfort level of non-uniform luminance distributions. *Misleading conclusions can be reached regarding comfort or glare comparisons between products, especially with clear sources and/or specular reflector surfaces.*

4.1 Circular Luminaire Openings (Open Bottom or Flat Lens)

For flat-lensed or open-bottom luminaires such as those illustrated in **Figure 4-1**, the equation for projected luminous area at angle θ is:

$$ABP = AB \cos \theta = \frac{\pi D^2 \cos \theta}{4}, \quad (4-4)$$

where:

ABP is the projected luminous area at angle θ

AB is the luminous area at 0 degrees

D is the diameter of the luminous area

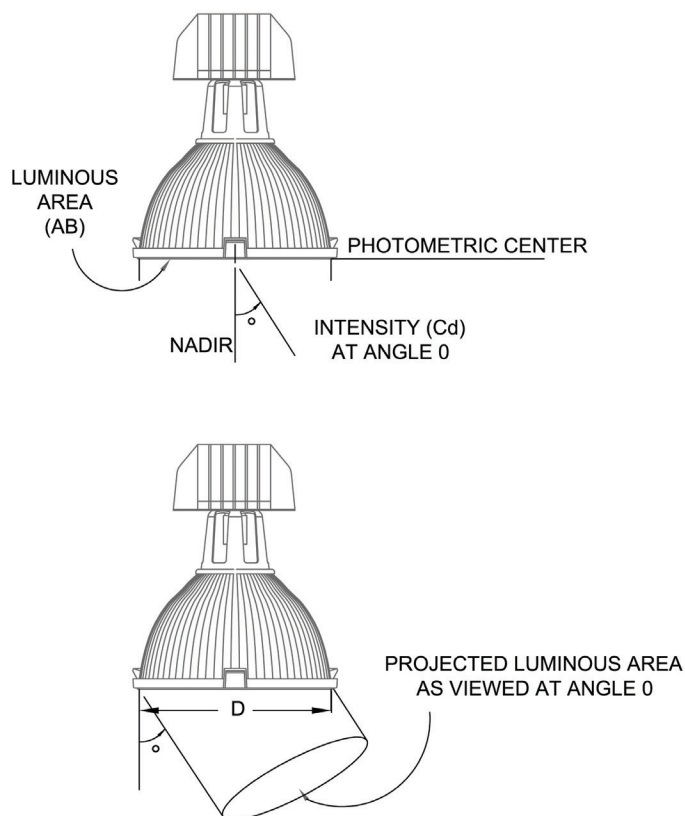


Figure 4-1. Luminous area and projected luminous area illustrated for flat-lens or open-bottom circular luminaires.
(© Illuminating Engineering Society)

4.2 Circular Luminaire Openings with Drop Lenses

For drop-lens circular luminaires (see **Figure 4-2**), the sides of the lens shall be considered in the total projected luminous area calculations. The total projected luminous area at angle θ equals $ASP + ABP$, where ASP is the projected side luminous area as viewed at angle θ (use appropriate formula for given geometry), and:

$$ABP = AB \cos \theta = \frac{\pi D^2 \cos \theta}{4}, \quad (4-5)$$

where:

ABP is the projected bottom luminous area as viewed at angle θ

AB is the bottom luminous area at 0 degrees

D is the diameter of the luminous area

A simple, straight (vertical) side drop lens would have a side projected area (ASP) of:

$$ASP = DH \sin \theta$$

4.3 Square or Rectangular Luminaire Openings (Open Bottom or Flat Lens)

For square or rectangular luminaire openings (see **Figure 4-3**), an appropriate projected area formula for a given geometry is used. Each unique situation shall be assessed and a faithful mathematical representation of the projected areas found. Candela values used in the luminance calculation shall be taken at the horizontal angle of interest. The area projection calculation shall be made at this same horizontal angle.

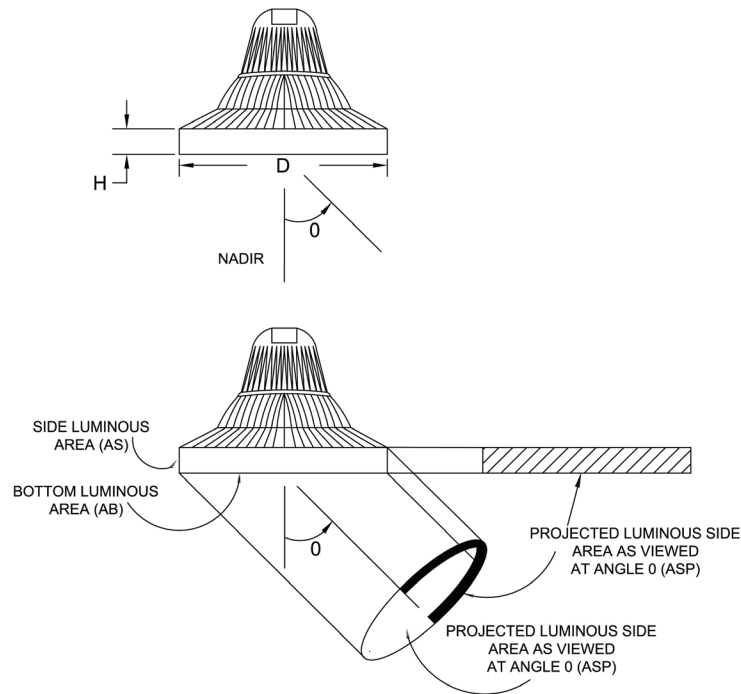


Figure 4-2. Bottom luminous area, side luminous area, and two projected luminous areas illustrated for drop-lens circular luminaires. (© Illuminating Engineering Society)

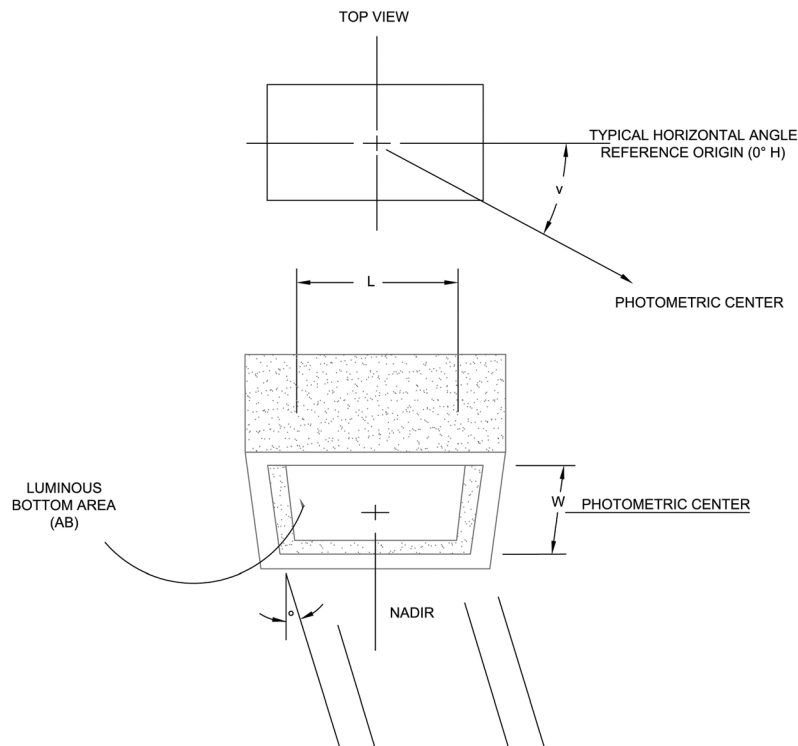


Figure 4-3. Bottom luminous area and projected luminous area illustrated for flat lens or open bottom square or rectangular luminaires. (© Illuminating Engineering Society)

4.4 Square or Rectangular Luminaire Openings with Drop Lenses

This case, of a square or rectangular luminaire opening with a drop lens (see **Figure 4-4**), is more complex than the flat-lens or open-bottom situation due to the fact that the amount of luminous side area in the drop portion of the lens changes with both horizontal and vertical viewing angles.

For a luminaire side (viewing the length dimension) **Equation 4-6** is used:

$$\text{total projected luminous side area} = ASP + ABP, \quad (4-6)$$

where:

ASP is the projected luminous side area (length) viewed at angle θ

ABP is the projected luminous bottom area (length) viewed at angle θ

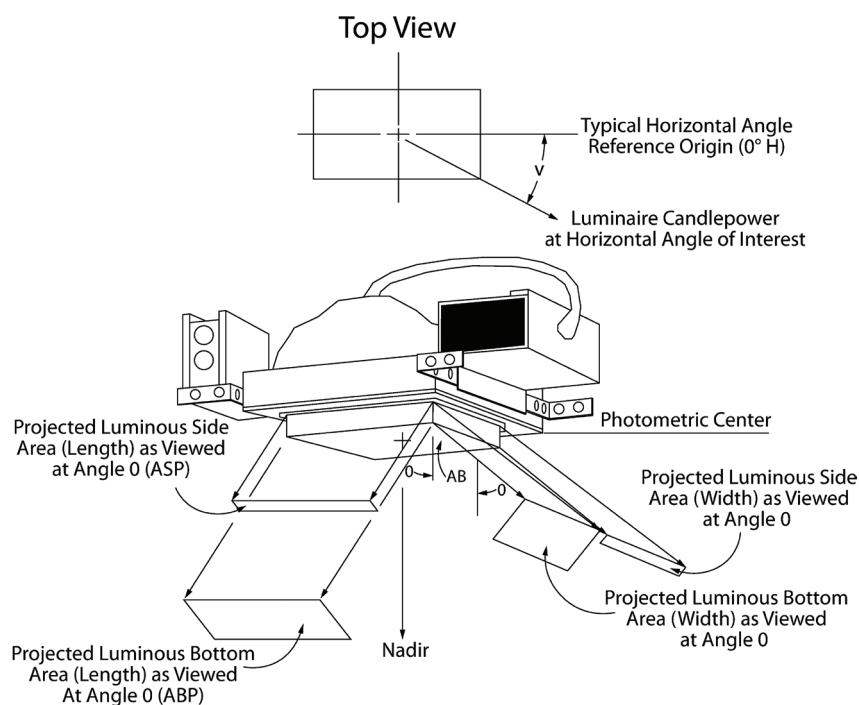


Figure 4-4. Bottom luminous area and four projected luminous areas illustrated for drop lens square or rectangular luminaires. (© Illuminating Engineering Society)

These area projections are calculated at "side" horizontal positions for given vertical angle " θ ". The total projected side area is used with the side candela value measured at angle θ to get the average side luminance of the luminaire.

For a luminaire end (viewing the width dimension) use the formula:

$$\text{total projected luminous end area} = ASP + ABP, \quad (4-7)$$

where:

ASP is the projected luminous end area (width) viewed at angle θ

ABP is the projected luminous bottom area (width) viewed at angle θ

The area calculations give different values when the luminous area is on a rectangular (non-square) luminaire. These area projections are calculated at "end" horizontal positions for given vertical angle θ . The total projected *end area* is used with the *end candela* value measured at angle θ to get average end luminance of the luminaire.

Average luminance calculated at other horizontal positions would require use of appropriate formulas to compute projected side and bottom areas at a given vertical angle θ and a given horizontal angle ψ .

4.5 Irregularly Shaped or Polygonal Openings for Luminaires

For flat-lensed or open-bottom luminaires having irregularly shaped or polygonal openings, the appropriate area formula is used to obtain projected area of luminous opening at a given horizontal position and vertical angle.

For an average luminance calculation at any horizontal viewing angle, the appropriate directional candela value at vertical angle θ is used. Calculated values are determined per **Section 4.0**.

For drop-lens versions of irregularly shaped luminaires, the bottom luminous area portion is computed using the appropriate equation for determination of the bottom area at nadir and then multiplying the area by $\cos\theta$. The projected side luminous area portion is determined for the geometry at hand with the appropriate formula.

The total projected luminous area at any given horizontal viewing angle is again $ASP + ABP$. Candela values are chosen to correspond to the horizontal position and vertical viewing angle of interest.

Extreme care should be exercised when calculating the projected area of these luminaires which have irregular shapes to ensure that accurate average luminance (calculated) values are obtained. Three-dimensional computer-aided design (CAD) programs may be utilized to accurately determine the projected areas.

Representative projected area formulas are provided in **Annex A** for convenience. However, these calculation examples should not limit the user's judgment in applying the most appropriate methods for the particular case at hand.

4.6 Projected Area Formula for Luminous Opening Types Defined in LM-63

The formulas shown below address specific luminous openings described in Annex D of *ANSI/IES LM-63-19, IES Standard File Format for the Electronic Transfer of Photometric Data and Related Information*.¹ Average luminance (calculated) is then determined from the base definition.

4.6.1 Point.

$$\text{Total projected luminous area: } N.A. \quad (4-8)$$

4.6.2 Rectangular.

$$\text{Total projected luminous area} = WL \cos\theta \quad (4-9)$$

4.6.3 Rectangular with Luminous Sides.

$$\text{Total projected luminous area} = WL \cos\theta + H \sin\theta [W(ABS)(\cos\psi) + L(ABS)(\sin\psi)] \quad (4-10)$$

4.6.4 Circular.

$$\text{Total projected luminous area} = \pi \left(\frac{D}{2}\right)^2 \cos\theta \quad (4-11)$$

4.6.5 Ellipse.

$$\text{Total projected luminous area} = \pi(W \cos\theta) \frac{L(ABS)}{4} \quad (4-12)$$

4.6.6 Vertical Cylinder.

$$\text{Total projected luminous area} = \pi \left(\frac{D}{2}\right)^2 \cos\theta + (ABS)(HD \sin\theta) \quad (4-13)$$

4.6.7 Sphere.

$$\text{Total projected luminous area} = \pi \left(\frac{D}{2} \right)^2 \quad (4-14)$$

4.6.8 Horizontal Cylinder along Photometric Horizontal.

$$YY = \sin\theta \cos\psi \quad (4-15)$$

$$VV = \left(\tan^{-1} \frac{YY}{\sqrt{1 - YY^2}} \right) \quad (4-16)$$

$$\text{Total projected luminous area} = \pi \left(\frac{D}{2} \right)^2 \sin VV + (ABS)(LD \cos VV) \quad (4-17)$$

4.6.9 Horizontal Cylinder Perpendicular to Photometric Horizontal.

$$XX = \sin\theta \cos\psi \quad (4-18)$$

$$VV = \left(\tan^{-1} \frac{XX \cos\theta}{\sqrt{1 - XX^2}} \right) \quad (4-19)$$

$$\text{Total projected luminous area} = \pi \left(\frac{D}{2} \right)^2 \sin VV + (ABS)(WD \cos VV) \quad (4-20)$$

4.6.10 Vertical Circle Facing Photometric Horizontal.

$$YY = \sin\theta \cos\psi \quad (4-21)$$

$$VV = ABS \left(\tan^{-1} \frac{YY}{\sqrt{1 - YY^2}} \right) \quad (4-22)$$

$$\text{Total projected luminous area} = \pi \left(\frac{D}{2} \right)^2 \sin VV \quad (4-23)$$

4.6.11 Vertical Ellipse Facing Photometric Horizontal.

$$YY = \sin\theta \cos\psi \quad (4-24)$$

$$VV = ABS \left(\tan^{-1} \frac{YY}{\sqrt{1 - YY^2}} \right) \quad (4-25)$$

$$\text{Total projected luminous area} = \pi \left(\frac{H}{2} \right) \left(\frac{D}{2} \right) \sin VV \quad (4-26)$$

4.7 Surfaces with Multiple Light Emitting Openings

Many luminaires are provided with light emitting surfaces which are composed of numerous openings, e.g., perforated or slotted areas.

The surface containing the openings is treated as the light emitting surface. The projected area for this surface is calculated using the equations described in this document for the particular surface. This area is then multiplied by the percent open area in this surface and becomes the projected luminous area for the equation below.

$$\text{Average luminance (calculated)} = \frac{\text{luminous intensity at angle } \theta}{\text{projected luminous area at angle } \theta} \quad (4-27)$$

This calculation method can be used with any of the area calculation methods described here or in **Annex A**, or applied to a calculated area obtained through three-dimensional computer-aided design (CAD) programs.

Annex A – Representative Area Formulas Reference (not all-inclusive)

The material in this Annex shows representative area formulas. It is intended to guide the user in adopting any appropriate method to compute projected areas. Originally, the equations presented in this Annex were derived from guidance provided in the *CRC Standard Mathematical Tables*, 21st ed., by the Chemical Rubber Co., 1973. The same information and mathematical guidance is available in Chapter 4 – Geometry, of the 33rd ed.² In this Annex, angles are defined as θ , representing horizontal angles, and ϕ , representing vertical angles.

A.1 Circular Luminaire Opening (Open Bottom or Flat Lens)

Viewing a circular flat lens from nadir produces the projected area seen in **Figure A-1**. This is a circle with the projected luminous area given by:

$$A = \frac{\pi D^2}{4} \quad (\text{A-1})$$

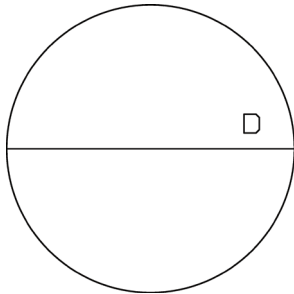


Figure A-1. A circular luminaire opening viewed from nadir (open bottom or flat lens).
(© Illuminating Engineering Society)

When this same circular flat lens is viewed at an angle of 60° from nadir, the result is an ellipse with the semi-major axis (A), the semi-minor axis (B), and the diameter of the circular opening (D), as seen in **Figure A-2**. The area is given by the following three equations. Therefore, the equations provided in **Section 4.1** of this document, do not need to be modified.

$$A = \frac{D}{2} \quad (\text{A-2})$$

$$B = A \cos \phi \quad (\text{A-3})$$

$$\text{Area} = \pi AB = \frac{\pi D^2 \cos \phi}{4} \quad (\text{A-4})$$

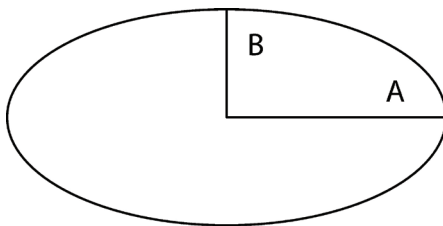


Figure A-2. A circular luminaire opening viewed from an angle of 60° (open bottom or flat lens). (© Illuminating Engineering Society)

A.2 Circular Luminaire Openings with Drop Lenses (Truncated Cone)

This example assumes a circular drop lens with a top diameter $D_1 = 60$ cm (2 ft), a bottom diameter $D_2 = 30$ cm (1 ft), and a height $H = 80$ cm (2.6 ft). **Figure A-3** shows such a truncated cone viewed from nadir. From this illustration, it is determined that the area seen is that of the top diameter only; therefore, the projected luminous area can be calculated from:

$$\text{Area} = \frac{\pi (D_1)^2}{4} \quad (\text{A-5})$$

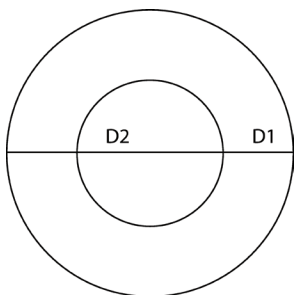


Figure A-3. A circular luminaire opening with a drop lens (truncated cone) viewed from nadir. (© Illuminating Engineering Society)

This is also seen to be true in **Figure A-4**, as the cone is rotated (up to a critical angle, ϕ_c). Beyond this critical angle (see **Figure A-5**), a portion of the top diameter, a portion of the bottom diameter, and a trapezoidal region between the two diameters is seen.

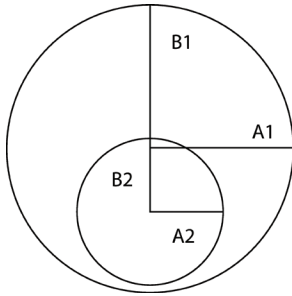


Figure A-4. A circular luminaire opening with a drop lens (truncated cone) viewed 10° from nadir. (© Illuminating Engineering Society)

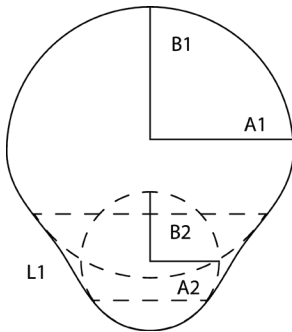


Figure A-5. A circular luminaire opening with a drop lens (truncated cone) viewed 20° from nadir. (© Illuminating Engineering Society)

Once the vertical angle reaches 90°, the method used to calculate area is the formula for the trapezoidal region as illustrated in **Figure A-6**:

$$Area = \frac{(D_1) + (D_2)}{2H} \quad (\text{A-6})$$

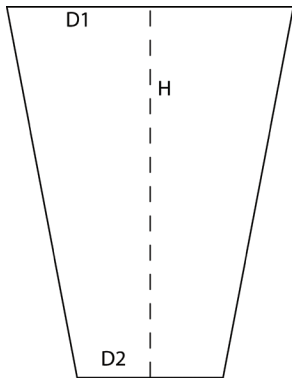


Figure A-6. A circular luminaire opening with a drop lens (truncated cone) viewed 90° from nadir. (© Illuminating Engineering Society)

Returning to the cases of partial cone rotation, it can be seen in **Figure A5** that the horizontal lines for the trapezoidal region do not coincide with the axes of the two ellipses representing the top and bottom surfaces. It can also be seen from **Figure A-4** and **Figure A-5** that the ellipse center for the bottom surface moves downward (and away) from the ellipse center for the top surface as the vertical angle increases. This is the same as rotating a point on the positive z -axis about the positive x -axis.³ From this, it can be seen that the critical angle (ϕ_c) occurs when the bottom edge of the smaller ellipse touches the bottom edge of the larger ellipse. The formulas for computing the area shown in **Figure A-4** (for vertical angles of ϕ between 0° and ϕ_c , inclusive) are:

$$A_1 = \frac{(D_1)}{2} \quad (\text{A-7})$$

$$B_1 = (A_1)\cos\phi \quad (\text{A-8})$$

$$A_1 = \frac{(D_2)}{2} \quad (\text{A-9})$$

$$B_2 = (A_2)\cos\phi \quad (\text{A-10})$$

$$\text{Area} = \pi(A_1)(B_1) = \frac{\pi(D_1)^2 \cos\phi}{4} \quad (\text{A-11})$$

When the vertical angle is greater than ϕ_c but less than 90° , the total projected area is composed of six component areas, with some points for the geometry as prescribed in **Figure A-5**. Assuming **Equations A-7, A-8, A-9, and A-10** still apply, then the six component areas that are used to calculate the total projected area for the case of **Figure A-5** can then be defined. Let the origin (0, 0) be at the center of the larger ellipse, and let the center of the smaller ellipse be at point $(0_{x2}, 0_{y2})$. Then $(0_{x2}, 0_{y2})$ is related to (0, 0) by the fact that the x-coordinate is 0 (zero) and the y-coordinate is $-H\sin\phi$, giving a coordinate value of $(0, -H\sin\phi)$. Let point (x_1, y_1) be the point of intersection of line L_1 and the larger ellipse [the intersection is in the third quadrant relative to the origin (0, 0)]. Let point (x_2, y_2) be the point of intersection of line L_1 and the smaller ellipse [the intersection is in the third quadrant relative to the origin $(0_{x2}, 0_{y2})$]. Line L_1 is the line that is tangent to both ellipses [the slope of the larger ellipse at point (x_1, y_1) is equal to the slope of the smaller ellipse at point (x_2, y_2) by the definition of tangency]. This diagram is symmetrical about the y-axis; therefore, the six areas of **Figure A-5** are defined as:

Area_1 = the upper half of the larger ellipse

Area_2 = the area under the larger ellipse between points $(-x_1, y_1)$ and (x_1, y_1) subtracted from the lower half of the larger ellipse

Area_3 = the rectangular area bounded by points $\{(-x_1, 0), (-x_1, -y_1), (x_1, -y_1), \text{ and } (x_1, 0)\}$

Area_4 = the area under the smaller ellipse between points $(-x_2, y_2)$ and (x_2, y_2)

Area_5 = the rectangular area bounded by the points $\{(-x_2, 0), (-x_2, -y_2), (x_2, -y_2), \text{ and } (x_2, 0)\}$

Area_6 = the trapezoidal region bounded by the points $\{(-x_1, y_1), (-x_2, -y_2), (x_2, -y_2), \text{ and } (x_1, -y_1)\}$

$$\text{Area}_1 = \frac{x(A_1)(B_1)}{2} \quad (\text{A-12})$$

$$\text{Area}_2 = \text{Area}_1 - \left[\frac{-(B_1)}{(A_1)} \right] \left[x_1 \sqrt{(A_1)^2 - (x_1)^2} + (A_1)^2 \sin^{-1} \left(\frac{x_1}{(A_1)} \right) \right] \quad (\text{A-13})$$

$$\text{Area}_3 = 2|x_1||y_1| \quad (\text{A-14})$$

$$\text{Area}_4 = \text{Area}_1 - \left[\frac{-(B_2)}{(A_2)} \right] \left[x_2 \sqrt{(A_2)^2 - (x_2)^2} + (A_2)^2 \sin^{-1} \left(\frac{x_2}{(A_2)} \right) \right] \quad (\text{A-15})$$

$$\text{Area}_5 = 2|x_2||y_2| \quad (\text{A-16})$$

$$\text{Area}_6 = \left[\frac{2x_1 + 2x_2}{2} \right] |y_2 - H \sin \phi + y_1| = [x_1 + x_2] |y_2 - H \sin \phi + y_1| \quad (\text{A-17})$$

$$\text{Area}_{\text{total}} = \text{Area}_1 + \text{Area}_2 + \text{Area}_3 + \text{Area}_4 - \text{Area}_5 + \text{Area}_6 \quad (\text{A-18})$$

Given the slope of a line tangent to the larger ellipse and the semi-major and semi-minor axis lengths, the point (x_1, y_1) can be determined with **Equations A-19 and A-20**:

$$x_1 = \frac{(A_1)^2 \tan \beta}{\sqrt{(B_1)^2 + (A_1)^2 \tan^2 \beta}} \quad (\text{A-19})$$

$$y_1 = -(B_1) \sqrt{1 - \frac{x_1^2}{(A_1)^2}} \quad (\text{A-20})$$

Similar equations can be developed for the smaller ellipse for point (x_2, y_2) :

$$x_2 = \frac{(A_2)^2 \tan \beta}{\sqrt{(B_2)^2 + (A_2)^2 \tan^2 \beta}} \quad (\text{A-21})$$

$$y_2 = -(B_2) \sqrt{1 - \frac{x_2^2}{(A_2)^2}} - H \sin \phi \quad (\text{A-22})$$

The values for x_1 and x_2 are negative when the slope is negative. The values for y_1 and y_2 are forced negative by taking the negative square root. Developing the equation for the line through point (x_2, y_2) yields:

$$y - y_2 = \tan \beta (x - x_2) \equiv y + H \sin \phi + (B_2) \sqrt{1 - \frac{(A_2)^2 \tan^2 \beta}{(B_2)^2 + (A_2)^2 \tan^2 \beta}} - (\tan \beta) x + \frac{(A_2)^2 \tan^2 \beta}{(B_2)^2 + (A_2)^2 \tan^2 \beta} = 0 \quad (\text{A-23})$$

The directed distance from point (x_1, y_1) to the line through (x_2, y_2) shall be zero for the two tangent lines to be the same line. Putting **Equation A.23** into the form $(Ax + By + C = 0)$ allows solving for the directed distance from a point $[P_1 = (x_1, y_1)]$ to the line through (x_2, y_2) using **Equation A.24**:

$$\frac{Ax_1 + By_1 + C}{\pm \sqrt{A^2 + B^2}} = s \quad (\text{A-24})$$

The directed distance from point (x_1, y_1) to the line through (x_2, y_2) is 0 when **Equation A-24** is 0 after substituting x_1 and y_1 for x and y in **Equation 4-15** (see **Section 4.6.8**). The result is an empirical equation that can be solved (for β) only by iterative techniques. Using the initial circular drop lens example having dimensions: top diameter $D_1 = 60$ cm (2 ft), bottom diameter $D_2 = 30$ cm (1 ft), height $H = 80$ cm (2.6 ft), and vertical angle $(\phi = 60^\circ)$, a graph can be obtained showing the relationship of directed distance (s) to slope (β) in radians (see **Figure A-7**). A gridded display is shown in **Figure A-8** (for approximation of the projected area).

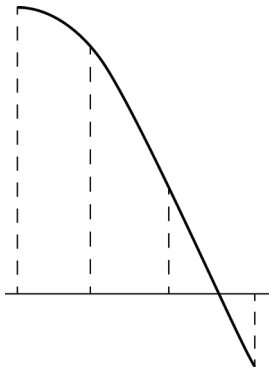


Figure A-7. The relationship between directed distances and angle of slope (β) in radians. (© Illuminating Engineering Society)

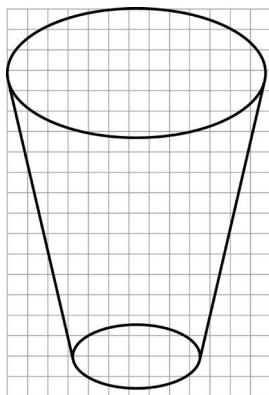


Figure A-8. This example of a circular luminaire opening with a drop lens (truncated cone) is viewed from 60° and includes a grid overlay for approximating the projected area. (© Illuminating Engineering Society)

A.3 Square or Rectangular Luminaire Openings (Open Bottom or Flat Lens)

The area of a polygon region bounded by straight line segments can be calculated from the determinant of the matrix of (x, y) coordinates of the vertices³; therefore, if the four corners of a rectangular flat lens are taken to be points P_1, P_2, P_3 , and P_4 , a matrix of (x, y, z) coordinates can be created for these four points and multiplied by two matrices (for the horizontal and vertical rotations) to determine the coordinate values to be used in the calculation of the area. An example is given in **Figure A-9** with $L = 60$ cm (2 ft) and $W = 30$ cm (1 ft). In **Figure A-9**, $\theta = 0^\circ$ and $\phi = 0^\circ$ (horizontal and vertical angles respectively), while in **Figure A-10** $\theta = 30^\circ$ and $\phi = 0^\circ$.

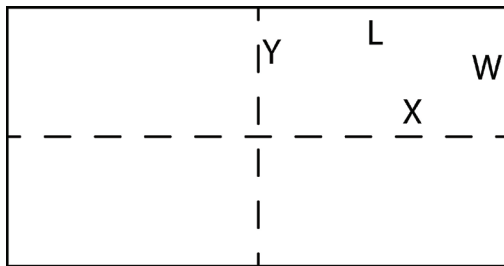


Figure A-9. A rectangular luminaire opening (open bottom or flat lens) prior to rotation where the horizontal angle $\theta = 0^\circ$ and the vertical angle $\phi = 0^\circ$. (© Illuminating Engineering Society)

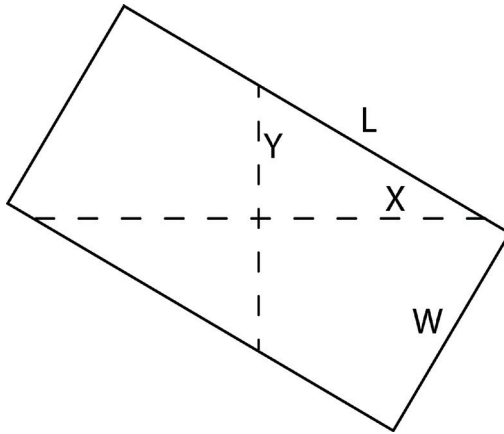


Figure A-10. A rectangular luminaire opening (open bottom or flat lens) during rotation, where the horizontal angle $\theta = 30^\circ$ and the vertical angle $\phi = 0^\circ$. (© Illuminating Engineering Society)

A horizontal rotation about the z -axis can be performed by matrix multiplication, whereby [the coordinates of the four corners before transformation] are multiplied by [the horizontal rotation about the z -axis], which equals [the transformed coordinates after horizontal rotation, the resultant matrix]. This process in mathematical form is:

$$\begin{bmatrix} \frac{-L}{2} & \frac{-W}{2} & 0 & 1 \\ \frac{L}{2} & \frac{-W}{2} & 0 & 1 \\ \frac{L}{2} & \frac{W}{2} & 0 & 1 \\ \frac{-L}{2} & \frac{W}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(-\theta) & \sin(-\theta) & 0 & 0 \\ -\sin(-\theta) & \cos(-\theta) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \left(\frac{-L}{2}\cos(-\theta) + \frac{W}{2}\sin(-\theta)\right) & \left(\frac{-L}{2}\sin(-\theta) - \frac{W}{2}\cos(-\theta)\right) & 0 & 1 \\ \left(\frac{L}{2}\cos(-\theta) + \frac{W}{2}\sin(-\theta)\right) & \left(\frac{L}{2}\sin(-\theta) - \frac{W}{2}\cos(-\theta)\right) & 0 & 1 \\ \left(\frac{L}{2}\cos(-\theta) - \frac{W}{2}\sin(-\theta)\right) & \left(\frac{L}{2}\sin(-\theta) + \frac{W}{2}\cos(-\theta)\right) & 0 & 1 \\ \left(\frac{-L}{2}\cos(-\theta) - \frac{W}{2}\sin(-\theta)\right) & \left(\frac{-L}{2}\sin(-\theta) + \frac{W}{2}\cos(-\theta)\right) & 0 & 1 \end{bmatrix}$$

(A-25)

The fourth dimension in the preceding is for scaling. The result of $\frac{1}{2}$ of the determinant of the (x, y) coordinates of the resultant matrix is length times width ($L \times W$), as expected. A vertical rotation about the x -axis (of the same initial matrix just presented) can be performed by matrix multiplication, whereby [the coordinates of the four corners before transformation] are multiplied by [the vertical rotation about the x -axis], which equals [the transformed coordinates after vertical rotation-the resultant matrix]. This process in mathematical form is:

$$\begin{bmatrix} \frac{-L}{2} & \frac{-W}{2} & 0 & 1 \\ \frac{L}{2} & \frac{-W}{2} & 0 & 1 \\ \frac{L}{2} & \frac{W}{2} & 0 & 1 \\ \frac{-L}{2} & \frac{W}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\phi & \sin\phi & 0 \\ 0 & -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{-L}{2} & \frac{W}{2}\cos\phi & \frac{-W}{2}\sin\phi & 1 \\ \frac{L}{2} & \frac{-W}{2}\cos\phi & \frac{-W}{2}\sin\phi & 1 \\ \frac{L}{2} & \frac{W}{2}\cos\phi & \frac{W}{2}\sin\phi & 1 \\ \frac{-L}{2} & \frac{W}{2}\cos\phi & \frac{W}{2}\sin\phi & 1 \end{bmatrix}$$

(A-26)

The result of $\frac{1}{2}$ of the determinant of the (x, y) coordinates of the resultant matrix in **Equation A-26** (the right-hand matrix) is length times width ($L \times W$) times the cosine of the vertical angle, as expected. Continuing this example for the case of a compound rotation (horizontal and vertical), the horizontal rotation matrix shall be applied first, then the vertical rotation matrix, to obtain the following resultant matrix:

$$\begin{bmatrix} \left(\frac{-L}{2}\cos(-\theta) + \frac{W}{2}\sin(-\theta)\right) & \left(\frac{-L}{2}\sin(-\theta) - \frac{W}{2}\cos(-\theta)\right)\cos\phi & \left(\frac{-L}{2}\sin(-\theta) - \frac{W}{2}\cos(-\theta)\right)\sin\phi & 1 \\ \left(\frac{L}{2}\cos(-\theta) + \frac{W}{2}\sin(-\theta)\right) & \left(\frac{L}{2}\sin(-\theta) - \frac{W}{2}\cos(-\theta)\right)\cos\phi & \left(\frac{L}{2}\sin(-\theta) - \frac{W}{2}\cos(-\theta)\right)\sin\phi & 1 \\ \left(\frac{L}{2}\cos(-\theta) - \frac{W}{2}\sin(-\theta)\right) & \left(\frac{L}{2}\sin(-\theta) + \frac{W}{2}\cos(-\theta)\right)\cos\phi & \left(\frac{L}{2}\sin(-\theta) + \frac{W}{2}\cos(-\theta)\right)\sin\phi & 1 \\ \left(\frac{-L}{2}\cos(-\theta) - \frac{W}{2}\sin(-\theta)\right) & \left(\frac{-L}{2}\sin(-\theta) + \frac{W}{2}\cos(-\theta)\right)\cos\phi & \left(\frac{-L}{2}\sin(-\theta) + \frac{W}{2}\cos(-\theta)\right)\sin\phi & 1 \end{bmatrix}$$

The result of $\frac{1}{2}$ of the determinant of the (x, y) coordinates of the preceding resultant matrix is also length times width ($L \times W$) times the cosine of the vertical angle. As in each previous example, the z coordinate values are made zero to project the points onto the x - y plane; therefore, the equations provided in **Section 4.4** of this document do not need to be modified.

A.4 Square or Rectangular Luminaire Openings with Drop Lenses

The same matrix formulas (as provided in **Section A.3**) can be used for the rectangular drop lenses. Each corner point (total of eight, see **Figure A-11**) is represented in the initial matrix and multiplied by the rotation matrices, and includes only those points that are in the outer boundary (after transformation) for inclusion in the calculation of the determinant. The points are gathered in a counterclockwise direction for calculation of the determinant, as was done in the examples of **Section A3**. It should be noted that only six of the eight corner points would be included in the calculation of the determinant for **Figure A-12**, since one corner is hidden behind the lens, and another corner (surrounded by lighted areas) is not on the boundary.

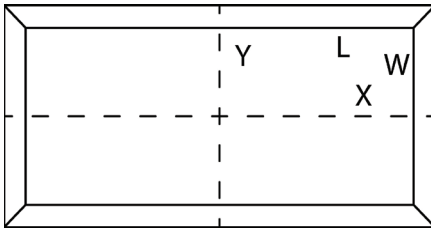


Figure A-11. Viewed from directly underneath (horizontal angle $\theta = 0^\circ$, vertical angle $\phi = 0^\circ$), a rectangular luminaire opening with a drop lens exhibits eight corner points that can be represented in the initial matrix. (© Illuminating Engineering Society)

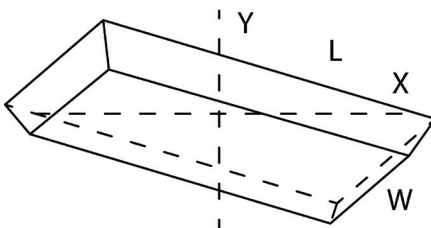


Figure A-12. After rotation (horizontal angle $\theta = 30^\circ$, vertical angle $\phi = 60^\circ$), the rectangular luminaire of Figure A-11 exhibits just six (of eight) corner points along its outer boundary. Only these six points are used in the calculation of the determinant for this particular view. (© Illuminating Engineering Society)

A.5 Spherical Lens

There are three cases for the spherical lens:

Case 1: The full sphere

Case 2: Less than a full sphere but at least half a sphere

Case 3: Less than half a sphere

For **Case 1**, the projected area is that of a circle, with the diameter of the circle equal to the diameter of the sphere.

For **Case 2**, the projected area is that of a circle (as in **Case 1**) up to a critical vertical angle (ϕ_c). The projected area is that of a partial circle and a partial ellipse once the vertical angle (ϕ) is greater than ϕ_c . **Figure A-13** shows **Case 2** viewed at an angle of 40° , with the diameter of the sphere $D = 30$ cm (12 in.) and the top 7.6 cm (3 in.) of the sphere removed. Below the dotted line (labeled "c"), the projected outline is an arc, and above it the projected outline is an ellipse. The projected area is that enclosed by the arc segment and the elliptical segment. When $0^\circ \leq \phi \leq \phi_c$, the projected area for **Case 2** is given by:

$$Area = \frac{\pi D_1^2}{4} \quad (A-27)$$

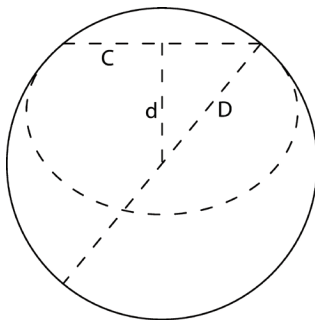


Figure A-13. A spherical lens that is less than a full sphere but at least half a sphere (Case 2) viewed at an angle of 40° . The top of the sphere (smaller portion) is removed. (© Illuminating Engineering Society)

However, when $\phi_c < \phi \leq 90$, the projected area calculation for **Case 2** becomes more complex, requiring that "arc" and "ellipse" components be computed:

$$H \equiv \text{Height} = D - \text{Amount Removed} \quad (A-28)$$

$$R = \frac{D}{2} \quad (A-29)$$

$$D_1 \equiv \text{Diameter at opening} \quad (A-30)$$

$$R_1 = \frac{D_1}{2} \quad (A-31)$$

$$\phi_c = \arctan\left(\frac{H - R}{R_1}\right) \quad (A-32)$$

$$d = \frac{H - R}{\sin\phi} \quad (A-33)$$

$$c = \sqrt{R^2 - d^2} \quad (A-34)$$

$$\phi = \arctan\left(\frac{c}{d}\right) \quad (A-35)$$

$$Area_{\text{arc}} = \pi R_1^2 + cd - R^2\theta \quad (A-36)$$

$$A_1 \equiv \text{semi-major axis} = \frac{D_1}{2} \quad (A-37)$$

$$B_1 \equiv \text{semi-minor axis} = R \cos(\phi - \phi_c) - (H - R) \cos(90 - \phi) \quad (A-38)$$

$$y_1 = \left(\frac{H - R}{\cos(90 - \phi)}\right) - (H - R) \cos(90 - \phi) \quad (A-39)$$

$$x_1 = A_1 \sqrt{1 - \frac{y_1^2}{B_1^2}} \quad (\text{A-40})$$

$$Area_{\text{ellipse}} = \left(\frac{B_1}{A_1} \right) \left[x_1 \sqrt{A_1^2 - x_1^2} + (A_1^2) \sin^{-1} \left(\frac{x_1}{A_1} \right) \right] - 2x_1 y_1 \quad (\text{A-41})$$

$$Area_{\text{total}} = Area_{\text{arc}} + Area_{\text{ellipse}} \quad (\text{A-42})$$

It should be noted in **Case 2** that once the vertical angle ϕ is 90° , the area of the ellipse is zero.

For **Case 3**, the projected area is that of a circle (with diameter equal to the opening) when the vertical angle ϕ is 0° . Under this unique condition, the projected area is given by:

$$Area = \frac{\pi D_1^2}{4} \quad (\text{A-43})$$

In all other instances for **Case 3**, the magnitude of the vertical angle (ϕ) is critical to determining the correct projected area. (**Figure A-14** shows less than half a sphere viewed from 60° .) When $0^\circ < \phi \leq \phi_c$, the projected area for **Case 3** is that of an ellipse:

$$R = \frac{D}{2} \quad (\text{A-44})$$

$$A_1 = \frac{D_1}{2} \quad (\text{A-45})$$

$$B_1 = R \cos \left[\phi + \tan^{-1} \left(\frac{R-H}{A_1} \right) \right] + (R-H) \cos(90 - \phi) \quad (\text{A-46})$$

$$Area = \pi A_1 B_1 \quad (\text{A-47})$$

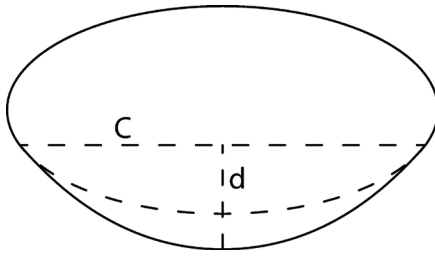


Figure A-14. A spherical lens that is less than a half sphere (Case 3) viewed at an angle of 60° . The top of the sphere (larger portion) is removed. (© Illuminating Engineering Society)

And when $\phi_c < \phi \leq 90^\circ$, the projected area for **Case 3** is that of an arc (partial circle) and a partial ellipse:

$$Area_{\text{arc}} = R^2 \phi - cd \quad (\text{A-48})$$

$$A_1 = \frac{D_1}{2} \quad (\text{A-49})$$

$$B_1 = R \cos \left(\phi + \tan^{-1} \left(\frac{R-H}{A_1} \right) \right) + (R-H) \cos(90 - \phi) \quad (\text{A-50})$$

$$y_1 = \left(\frac{R-H}{\cos(90 - \phi)} \right) - (R-H) \cos(90 - \phi) \quad (\text{A-51})$$

$$x_1 = A_1 \sqrt{1 - \frac{y_1^2}{B_1^2}} \quad (\text{A-52})$$

$$Area_{\text{ellipse}} = \pi A_1 B_1 + 2x_1 y_1 - \left(\frac{B_1}{A_1} \right) \left[x_1 \sqrt{A_1^2 - x_1^2} + (A_1^2) \sin^{-1} \left(\frac{x_1}{A_1} \right) \right] \quad (\text{A-53})$$

$$Area_{\text{total}} = Area_{\text{arc}} + Area_{\text{ellipse}} \quad (\text{A-54})$$

In **Case 3** the projected area for the elliptical portion is zero once the vertical angle (ϕ) is 90° .

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